SE Lab 3

SRN: PES1UG24AM278 | SECTION: E

Architectural Justification

Academic Elective Bidding & Allocation System

We chose Layered Architecture (Presentation → Business Logic → Data) for the Academic Elective Bidding & Allocation System.

Architectural Choice

A three-tier Layered Architecture, separating the Presentation layer (Student Portal, Registrar Console), the Business Logic layer (Bid Management Service, Elective Catalog Service, Allocation Solver Engine, Auth & Access Control Service), and the Data layer (Academic & Bidding Data Store).

Reason 1 – Clean separation of independently-changing concerns

The system naturally splits into UI-facing workflows (browsing electives, ranking preferences, registrar configuration), rule-driven logic (credit-limit validation, prerequisite checks, the allocation solver), and persisted records (student academic history, bids, elective catalog). Layering lets the allocation algorithm in FR-005 be revised or re-tuned without touching the student/registrar UI or the database schema, and lets elective catalog rules (FR-004) change each semester without redeploying the bidding UI.

Reason 2 – Fit for a single-institution, cycle-based workload

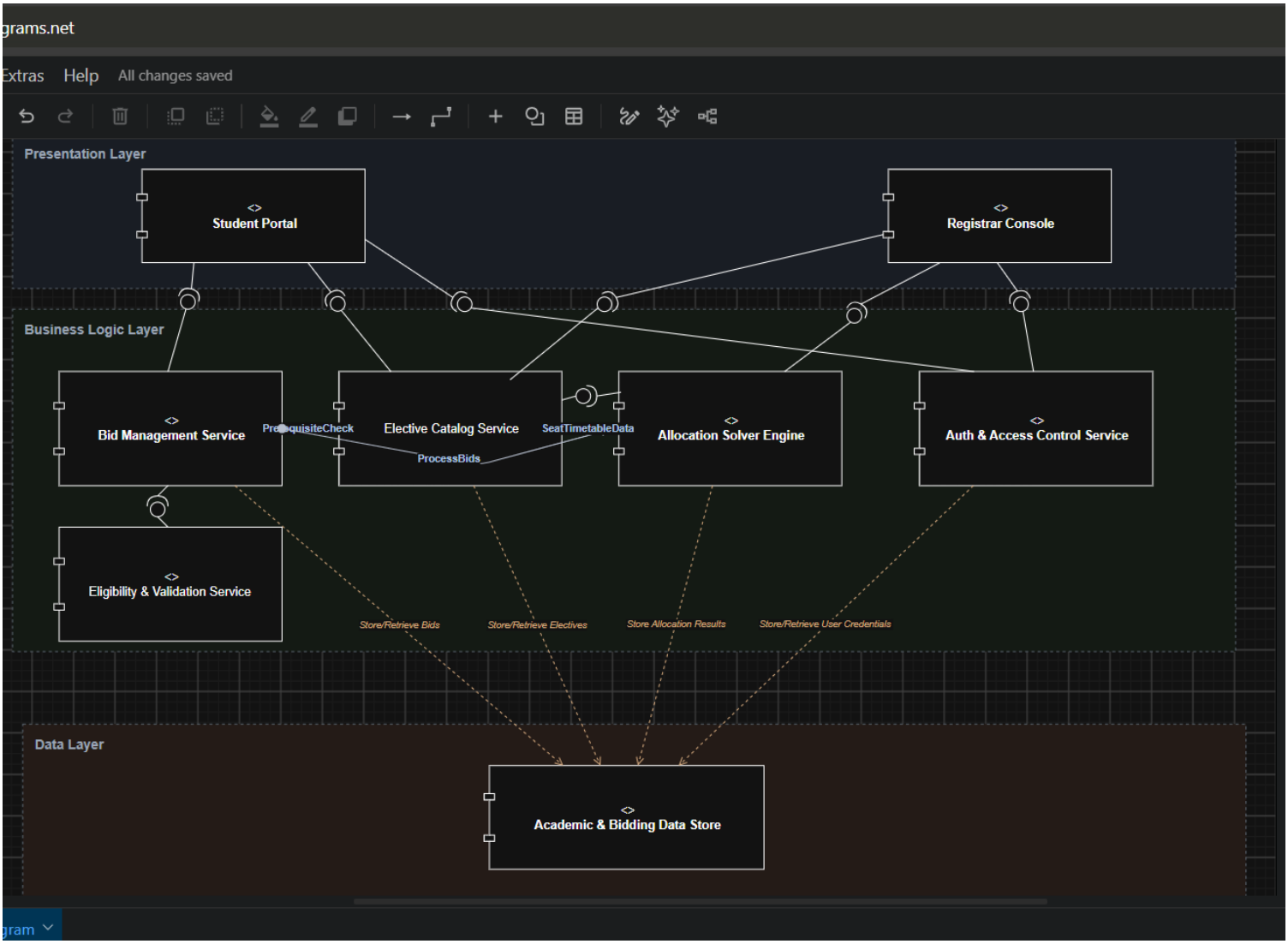
Bidding runs in fixed windows followed by one allocation run per cycle (FR-001–FR-003), not a continuously-scaling, multi-tenant workload. A layered structure is far simpler to build, secure, and operate at this scale than a distributed Microservices style (which adds network latency and operational overhead the allocation solver cannot afford) or a plain Client-Server split (which does not separate business rules from data access cleanly enough to protect prerequisite and capacity logic).

Security Advantage

All requests from the Student Portal and Registrar Console pass through the Auth & Access Control Service in the business layer before reaching any business logic or data component. This single choke point enforces role-based access (NFR-002): students can never invoke registrar-only operations such as configuring electives or running the allocation solver, and the Data layer is never reached directly by an unauthenticated request.

Performance Benefit

Because the Allocation Solver Engine sits in the same business layer as the Bid Management and Elective Catalog services, it reads bids, seat capacities, prerequisites, and timetable data through fast in-process/local calls rather than remote network calls between independently deployed services. This keeps end-to-end allocation time low, supporting the NFR-001 target of processing 5,000 student bids and resolving schedule conflicts in under 30 seconds.


grams.net
Extras
Help
All changes saved
Presentation Layer
Student Portal
Registrar Console
Business Logic Layer
Bid Management Service
PrerequisiteCheck
Elective Catalog Service
SeatTimetableData
Allocation Solver Engine
Auth & Access Control Service
ProcessBids
Eligibility & Validation Service
Store/Retrieve Bids
Store/Retrieve Electives
Store Allocation Results
Store/Retrieve User Credentials
Data Layer
Academic & Bidding Data Store
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